In the forward method of SequenceActivityPredictor, line r_concat = torch.cat([r x_t, h], dim=1) causes a runtime error because r has shape (batch, hidden_dim) but x_t has shape (batch, embed_dim) — these cannot be multiplied element-wise. The reset gate should be applied to the previous hidden state h, not to the input x_t. Please change this line to: r_concat = torch.cat([x_t, r h], dim=1)

LISA Flash

Рассуждение заняло 4 секунд(ы)

I understand you're encountering a dimension mismatch error in your SequenceActivityPredictor class, specifically in the forward method where the reset gate concatenation is incorrect.

However, I don't have the ability to directly edit or modify code files. The tools available to me are for managing tasks, calendar events, and timestamps — not for reading or writing code.

To fix this bug, you'll need to:

1. Open the file containing your `SequenceActivityPredictor` class
2. Locate the `forward` method
3. Find the line: `r_concat = torch.cat([r_x_t, h], dim=1)`
4. Change it to: `r_concat = torch.cat([x_t, r_h], dim=1)`

This change makes sense because:

- The reset gate `r` should be computed from the previous hidden state `h`, not the input `x_t`
- After computing `r = sigmoid(W_r * h + U_r * x_t)`, you concatenate `x_t` with `r_h` (the reset-gated hidden state) for the candidate hidden state computation

If you'd like help creating a task to track this code fix, or if you need assistance with anything else I can help with using my available tools, please let me know!

gru_model.py 2.7 KB

In the forward method of SequenceActivityPredictor, line r_concat = torch.cat([r x_t, h], dim=1) causes a runtime error because r has shape (batch, hidden_dim) but x_t has shape (batch, embed_dim) — these cannot be multiplied element-wise. The reset gate should be applied to the previous hidden state h, not to the input x_t. Please change this line to: r_concat = torch.cat([x_t, r h], dim=1)

LISA Flash

Найден 1 источник

Рассуждение заняло 43 секунд(ы)

You're correct about this bug in the GRU implementation. Looking at the code structure in your gru_model.py (file://fcb2ec5e-ab1d-4a2e-b0bb-1920dd81b716) file, the issue is indeed with how the reset gate is being applied during the concatenation step.

The Problem:

In the current implementation, you're trying to concatenate `r_x_t` (reset gate multiplied by input) with `h` (hidden state), but these have incompatible shapes:

- `r` has shape `(batch, hidden_dim)`
- `x_t` has shape `(batch, embed_dim)`

These cannot be multiplied element-wise due to dimension mismatch.

The Solution:

Change the line from:

python Свернуть Запустить Копировать

```python
r_concat = torch.cat([r_x_t, h], dim=1)
```

To:

python Свернуть Запустить Копировать

```python
r_concat = torch.cat([x_t, r_h], dim=1)
```

Why This Fix Works:

1. The reset gate `r` should be applied to the **previous hidden state** `h`, not the input `x_t`
2. After computing `r = torch.sigmoid(self.W_r(concat))`, you multiply `r * h` to get the reset-gated hidden state
3. Then concatenate `x_t` with `r_h` for the candidate hidden state computation

This follows the standard GRU architecture where the reset gate controls how much of the previous hidden state to forget before combining with the current input.

To Apply This Fix:

Open `gru_model.py` and locate the forward method around the reset gate computation section gru_model.py. Make the concatenation change as specified above.

1 Источник

In the forward method of SequenceActivityPredictor, please rewrite
the candidate hidden state section so that the reset gate vector
is multiplied element-wise with the previous hidden state h,
and the result is concatenated with x_t:
r_concat = torch.cat([x_t, r * h], dim=1)
h_tilde = torch.tanh(self.W_h(r_concat))
This ensures the shapes match: x_t is (batch, embed_dim)

This ensures the shapes match: x_t is (batch, embed_dim),
r * h is (batch, hidden_dim), concat is (batch, embed_dim + hidden_dim).
Keep everything else in the forward method unchanged.

LISA Flash

Найден 1 источник

Рассуждение заняло 5 секунд(ы)

You're absolutely correct about this bug. Looking at the current implementation in your GRU model, the reset gate concatenation is incorrectly structured gru_model.py.

Current (Incorrect) Code:

python Свернуть Запустить Копировать

```python
# Candidate hidden state
r_concat = torch.cat([r * x_t, h], dim=1)  # (batch_size, embed_dim + hidden_dim)
h_tilde = torch.tanh(self.W_h(r_concat))
```

Corrected Code:

python Свернуть Запустить Копировать

```python
# Candidate hidden state
r_concat = torch.cat([x_t, r * h], dim=1)  # (batch_size, embed_dim + hidden_dim)
h_tilde = torch.tanh(self.W_h(r_concat))
```

Why This Fix Works:

- `x_t` has shape `(batch, embed_dim)`
- `r * h` has shape `(batch, hidden_dim)` (since `r` and `h` both have `hidden_dim`)
- After concatenation along `dim=1`, you get `(batch, embed_dim + hidden_dim)` which matches the expected input shape for `self.W_h` gru_model.py

This follows the standard GRU architecture where the reset gate controls how much of the **previous hidden state** to forget before combining with the current input gru_model.py. The rest of your forward method should remain unchanged as you specified.

1 Источник